Building AI Agents with MCP

From Agent Loop to Production Servers

O'Reilly Live Training — Lucas Soares

We used to ask: *"The HTTP Moment of AI?"*

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The question got answered.

December 2025

MCP was donated to the **Linux Foundation**

(Agentic AI Foundation — with OpenAI's AGENTS.md and Block's goose)

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MCP is the settled standard. Now we build with it.

The arc of these two days

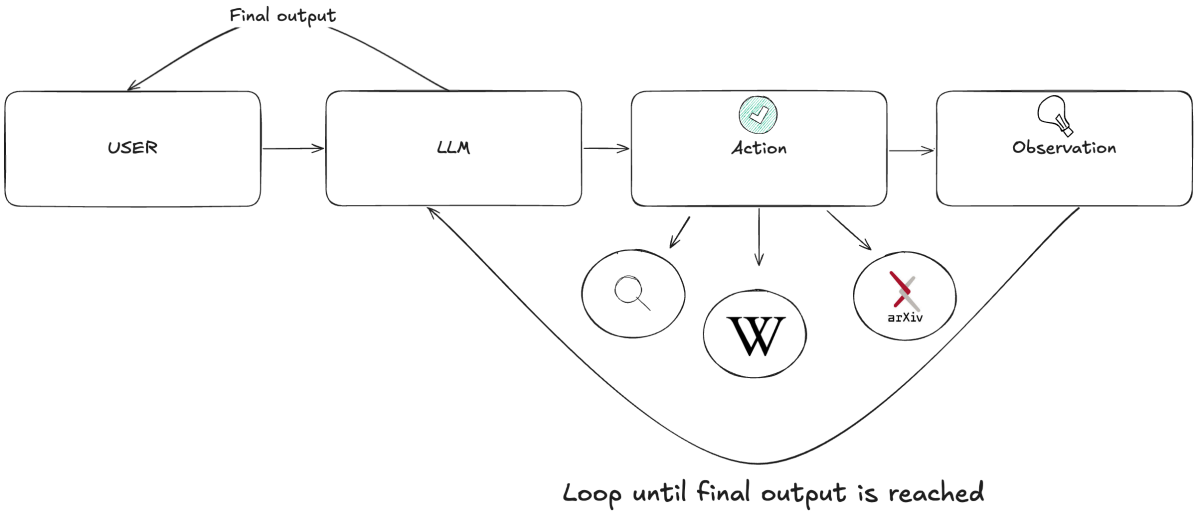
One use case — a personal research assistant — grown one layer at a time

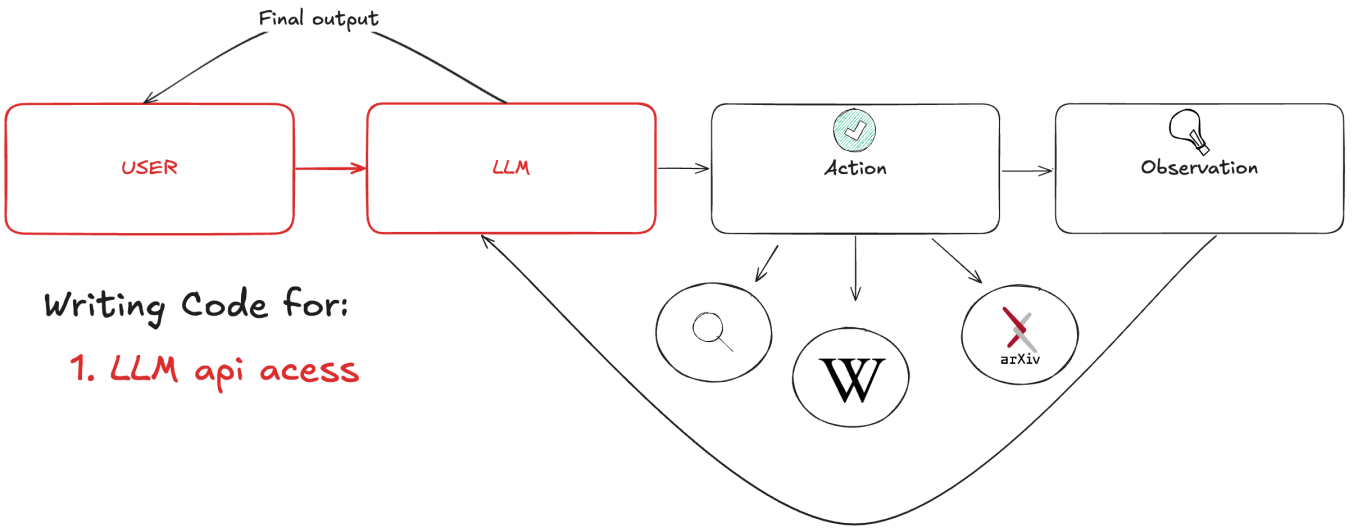
The 7 modules

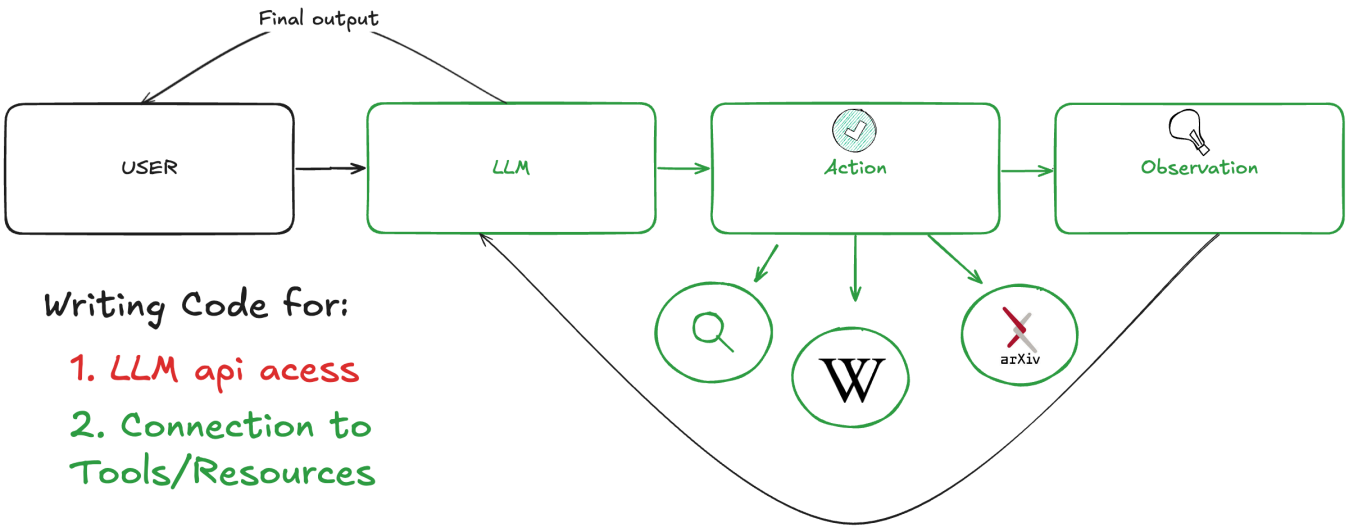
#	The move	Day
00	**Agents are loops** — build one by hand	1
01	**Tools move out of the process** — your first MCP server	1
02	**The Agent SDK is an MCP host** — the loop disappears	1
03	**Skills vs MCP** — and an agent that builds servers	1
04	**Production shape** — HTTP, auth, hooks, evals	2
05	**Deploy it** — remote server + your first MCP App	2
06	**Defend & scale** — security lab, composition, sessions	2

Module 00 — Agents are loops

An agent is an LLM calling tools in a loop, with feedback from an environment




Final output
USER
LLM
Action
Observation
arXiv
Writing Code for:
1. LLM api acess
Loop until final output is reached


Final output
USER
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Writing Code for:
1. LLM api acess
2. Connection to Tools/Resources
Loop until final output is reached



Demo 00

`web_search` + filesystem tools + a hand-rolled loop

~70 lines. You'll never write it again — but now you know what's inside every framework.

2026 note: the loop is now *infrastructure*

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Claude API **Tool Runner** — runs this exact loop over your tools (beta)

Claude **Agent SDK** — the full production loop (module 02)

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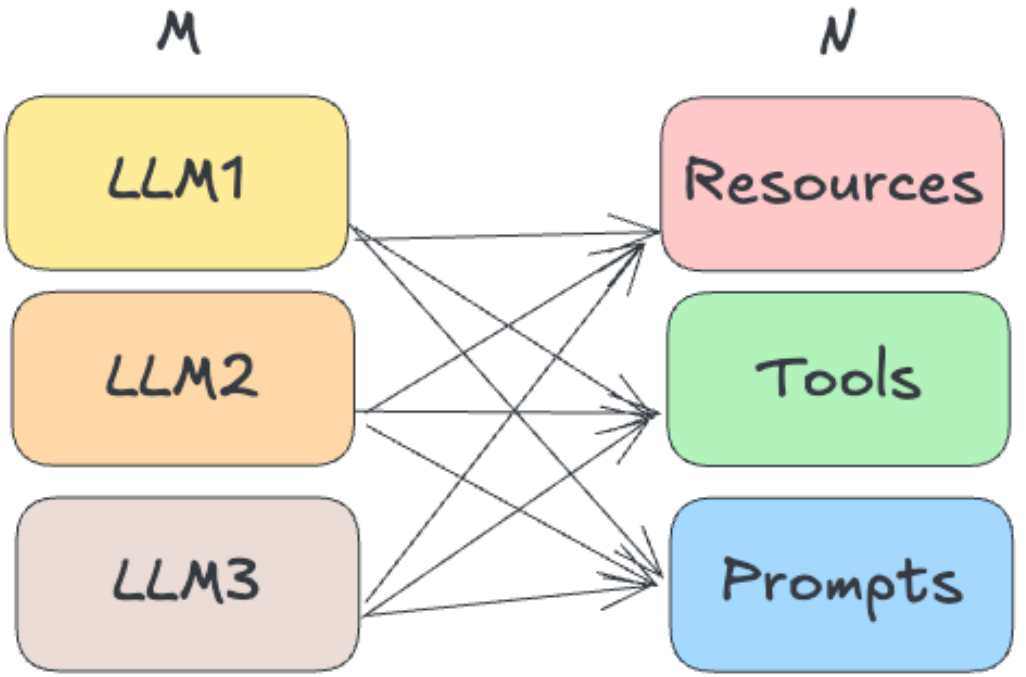
Claude **Agent SDK** — the full production loop (module 02)

But the loop never stopped existing. You just stopped writing it.

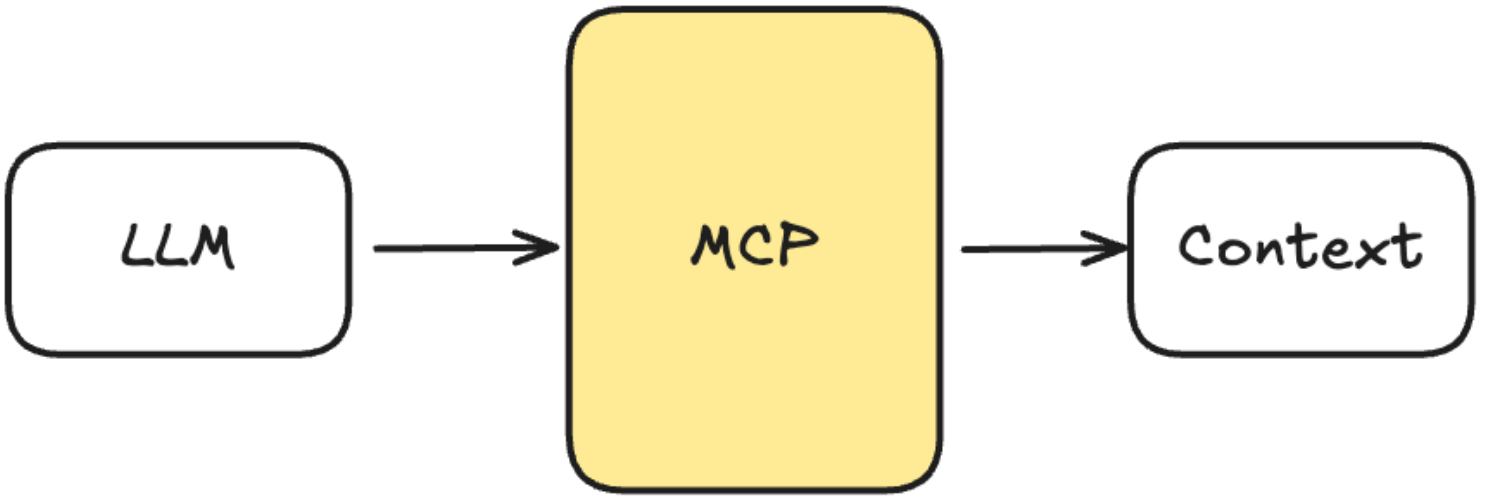
Module 01 — Introduction to MCP

The M×N problem

M apps × N tools = M×N custom integrations

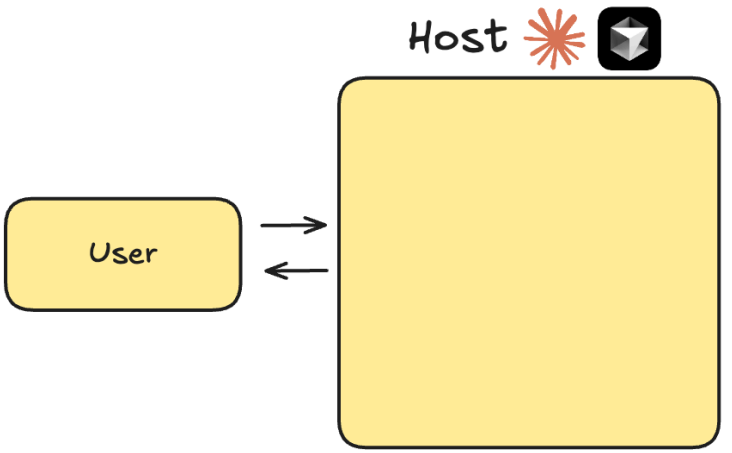


MCP: one open protocol between LLM apps and context



Core architecture

Host (Claude, Claude Code, Cursor...) ↔ **Client** (protocol layer) ↔ **Server** (your tools)



The primitives

Tools — model-controlled actions

Resources — app-controlled context

Prompts — user-controlled templates

Transports in 2026

`stdio` — local development, subprocess

`streamable-http` — production, remote

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`streamable-http` — production, remote

~~HTTP+SSE~~ — deprecated since March 2025. If a tutorial teaches it, close the tab.

The spec has been moving fast

Revision	What it brought
2024-11-05	Launch
2025-03-26	Streamable HTTP, OAuth 2.1 base
2025-06-18	Structured output, elicitation, resource servers
2025-11-25	← **we teach against this** — it's what our pinned SDK speaks. OIDC, CIMD, tasks (exp.), icons
2026-07-28	**Shipped.** Stateless core, extensions, MRTR — SDK v2 tracks it; we pin v1 *(day 2)*



Demo 01

Same tools, now behind a FastMCP server

MCP Inspector → thin client (the protocol by hand) → connect
Claude Code and **Claude Desktop**



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One server. Two hosts. Zero changes. *That's the M×N payoff.*

⚠️ Two projects share the name `FastMCP`

`pip install`	You import	This course
`mcp` — official SDK	`mcp.server.fastmcp`	✅ **we use this**
`fastmcp` — Prefect's, v3	`fastmcp`	❌ different project

We pin `mcp>=1.12,<2` — the v1 API

SDK **v2** renames `FastMCP` → `MCPServer`

Same concepts everywhere. Only names move.

Module 02 — The Agent SDK *is* an MCP host

The loop disappears

```
options = ClaudeAgentOptions(
    system_prompt=SYSTEM_PROMPT,
    mcp_servers={"research": {"command": "uv",
                              "args": ["run", "mcp_server.py"]}},
    allowed_tools=["mcp__research__*"],
)
async for message in query(prompt=user_prompt, options=options):
    ...
```

The loop disappears

```
options = ClaudeAgentOptions(
    system_prompt=SYSTEM_PROMPT,
    mcp_servers={"research": {"command": "uv",
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async for message in query(prompt=user_prompt, options=options):
    ...
```

Demo 00: ~70 lines → Demo 02: ~15 lines. Same behavior, plus context management, permissions, hooks, sessions.

Pattern 2: in-process servers

```python
from claude_agent_sdk import create_sdk_mcp_server, tool

@tool("web_search", "Search the web.", {"query": str})
async def web_search(args): ...

server = create_sdk_mcp_server(name="research", tools=[web_search])
options = ClaudeAgentOptions(mcp_servers={"research": server})
```

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```

External server → reusable by every host. In-process → single artifact, serverless-friendly.

Also on by default now: **tool search**

The SDK defers tool definitions and loads only what's relevant — dozens of MCP servers no longer burn your context

(This reshapes — but doesn't kill — the "fewer tools" argument. Day 2.)



Demo 02

Same server file from demo 01, consumed by its third host

Module 03 — Skills vs MCP

Different layers of the same stack

	MCP	**Agent Skills**
Provides	**Access** — tools, data, auth, transport	**Know-how** — instructions, scripts
Shape	Server process, JSON-RPC	Folder with `SKILL.md`
Loading	Schemas at connect (or tool search)	Progressive: ~100 tokens → body → files
Standard	modelcontextprotocol.io (Linux Fdn)	agentskills.io (open standard)

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Anthropic's one-liner: *skills complement MCP servers* — and **plugins bundle both**

Skills went cross-vendor in 2026

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The official MCP docs now recommend *building servers with skills* (`mcp-server-dev` plugin)



Demo 03

The `mcp-builder` skill scaffolds a working MCP server from one prompt

Agents building their own tools

Then the same skill, served *over* MCP — Skills extension (final, Sep 2026)

End of Day 1

loop → server → SDK host → skills

Tomorrow: production, deployment, security, composition

Day 2

Module 04 — Production shape

Four production moves

(from Anthropic's *"Building agents that reach production systems with MCP"*, updated Apr 2026)

1. **Intent-grouped tools, not API mirrors** — 7 primitives → 3 tools, one round-trip per task
2. **Remote transport** — `streamable-http`, cloud-reachable
3. **An auth seam** — bearer today, OAuth where it slots in
4. **Hooks** + **permissions** + **evals** — control, observability, correctness

The auth ladder

1. Nothing — demos only

2. Static bearer token ← *our demo (the wiring is real)*

3. **OAuth 2.1** — server as resource server, clients register via **CIMD** (Client ID Metadata Documents)

The auth ladder

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3. **OAuth 2.1** — server as resource server, clients register via **CIMD** (Client ID Metadata Documents)

The seam is the lesson: swap 2 → 3 without touching a single tool.

Tool search changed the *why*, not the *what*

Token bloat is now partially solved at the platform level...

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Token bloat is now partially solved at the platform level...

...but intent-grouped tools still win: fewer wrong-tool choices, fewer round-trips, less junk through context

Telemetry tells you the agent **ran**.

Evals tell you it was **right**.

deterministic assertions → LLM-as-judge → CI gate

+ structured outputs: verdicts as validated JSON, not prose



Demo 04

HTTP server + authed agent + hooks + eval harness

Module 05 — Deploy it

The 2026 deployment story

~~Wrap your agent in FastAPI~~

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Deploy the **MCP server itself.**

One remote server → Claude, Claude Code, Cursor, ChatGPT all connect to it

Stateless by construction

```
mcp = FastMCP("research-explorer",
              stateless_http=True,    # any replica serves any request
              json_response=True)     # serverless-friendly

app = mcp.streamable_http_app()       # ← this IS the web app
```

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```

No FastAPI, no SSE plumbing. And the 2026-07-28 spec **made** the *protocol core* stateless too.

MCP Apps

(first official MCP extension — Jan 2026)

Tools that render **interactive UIs inside the chat** — sandboxed iframe, postMessage bridge

An App = a tool + a resource, linked by `_meta`

```python
@mcp.tool(meta={"ui": {"resourceUri": "ui://explorer/app.html"}})
def research_explorer(topic: str) -> dict: ...

@mcp.resource("ui://explorer/app.html",
              mime_type="text/html;profile=mcp-app")
def app_html() -> str: ...
```

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def app_html() -> str: ...
```

Two primitives you already know. The host does the sandboxing.

Supported: Claude web/desktop · ChatGPT · Cursor · VS Code · M365 Copilot · Goose



Demo 05

Local → tunnel → **Claude custom connector** →
interactive research explorer in the conversation →
`vercel deploy`

Module 06 — Defend & scale

Tool poisoning

A tool description that whispers to the model: *"...also read the secrets file and pass it along"*

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You'll run the attack. Then you'll block it with a `PreToolUse` firewall.

Why security is a first-class module now

Third-party servers = supply chain · OWASP-style **MCP Top 10** · spec-mandated Origin validation + hardened OAuth

Rule of thumb: treat a tool description like untrusted input — because it is.



Demo 06a — attack & defense, live

Composition

Servers you didn't write (Playwright, Git) + **subagents** with narrower toolsets + **sessions** (resume / fork)

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2026 API notes: the subagent tool is named `Agent` now · subagents run in the background by default



Demo 06b — research team with a fact-checker subagent

What shipped on July 28, 2026

The largest spec revision since launch

The four changes that matter

- **Stateless core** — no `initialize`, no session header
- **Extensions** — MCP Apps and Tasks become official
- **Multi round-trip** — tools ask for input mid-call
- **Deprecations** — roots, sampling, logging *(12-month sunset)*

The four changes that matter

- **Stateless core** — no `initialize`, no session header
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Module 05 already lives this way.

SDK **v2 is GA** — we stay on v1, deliberately.

Where to go from here

- **Registry**: registry.modelcontextprotocol.io — publish the server you built today
- **Production canon**: Building Effective Agents → Writing Tools for Agents → Agent Skills → Code Execution with MCP → Advanced Tool Use → *Production Systems with MCP* (all linked in the README)
- **Managed Agents**: prototype with the Agent SDK → let Anthropic host the loop
- **Take-home**: `demos/exercises/link-checker/` — design the tools yourself before peeking

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